Riccardo **Casciotti**

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Summary

Computer Science Engineer and Doctoral Researcher at Tampere University, specializing in Continual Learning and Deep Learning for sound classification. My research focuses on understanding and mitigating catastrophic forgetting in neural networks, investigating how learned representations evolve across sequential learning tasks and exploring methods to improve knowledge retention and model adaptability.
My background encompasses Computer Vision, time-series forecasting, and statistical analysis, with experience in developing, training, and evaluating deep learning models using high-performance computing infrastructure. I have a strong interest in the mathematical foundations of machine learning, neural network interpretability, and the development of novel learning methodologies.

Work Experience

Tampere University — *Tampere, Finland*

Doctoral Researcher, Continual Learning for Sound — *May. 2025 - present*

- Doctoral researcher in The Machine Listening group focused on Continual Learning techniques for Audio.
- Research catastrophic forgetting in class-incremental sound classification, studying retention of earlier knowledge as models learn new tasks.
- Design and evaluate continual learning experiments, including comparisons of model representations and activation patterns across training stages.
- Develop and run deep learning training and evaluation workflows with PyTorch on HPC infrastructure using SLURM.
- Contribute to research publications and the DCASE 2026 challenge on domain-agnostic incremental sound classification.
- Teacher Assistant in the Advance Audio Processing course held at Tampere University.

Superprof — *Milan, Italy*

Private Teacher, ∼ 100 hours of class — *Jan. 2020 - Jun. 2024*

- Teacher to university students of algorithms, data structures, programming languages and technologies for the web, programming languages.
- English teacher to high school and university students.

LUISS Guido Carli — *Rome, Italy*

Software Architect, Phd Assistant — *Dec. 2020-Jan. 2021*

- Projected and implemented a Web Scraper service which gathered government data about a country's economy.
- The service calculates statistics about the data gathered and provides reports files.

Skills

Research	**Continual learning, catastrophic forgetting, representation analysis, sound classification, deep learning**
Machine Learning	**NumPy, Pandas, Ray, PyTorch, Keras, TensorFlow, Matlab, Scikit-Learn, Matplotlib**
DevOps	**AWS, Docker, Kubernetes, High Performance Computing, SLURM**
Back-end	PostGreSQL, Node.js, Express.js, Vue.js, REST API, OAuth
Front-end	Vue.js, HTML5, CSS
Programming	**C, Python, Java**, JavaScript, SQL, Scala, Assembly, php, Dart, bash
Soft-Skills	Time Management, Problem Solving, Project Analysis, Team work, Communication skills
Languages	Italian, **English (IELTS: 7.5/9)**

Publications

Investigating catastrophic forgetting in sound event classification

Riccardo Casciotti, Annamaria Mesaros

IEEE International Workshop on Multimedia Signal Processing (MMSP), 2026

Domain-Agnostic Incremental Learning for Sound Classification: A DCASE 2026 Challenge Task

Riccardo Casciotti, Manjunath Mulimani, Manu Harju, Jesper Rindom Jensen, Annamaria Mesaros

DCASE 2026

Incremental learning for audio classification with Hebbian Deep Neural Networks

Riccardo Casciotti, Francesco De Santis, Alberto Antonietti, Annamaria Mesaros

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026

Education

Tampere University
Tampere, Finland

PhD in Continual Learning for Sound
2025 - Ongoing

- Doctoral Researcher with specialization in regularization techniques, architectural techniques, replay solutions, and explainability in deep neural networks aimed at reducing and analyzing catastrophic forgetting and balancing the memory retention-plasticity tradeoff.

Politecnico di Milano
Milan, Italy

MSc in Computer Science and Engineering
2021 - 2025

- Master's Thesis on bio plausible approaches to implement Continual Learning in a bio inspired Deep Neural Network for Image Classification.
- Specialization on Artificial Intelligence, Deep Learning and Wireless Communications.

Università di Roma Sapienza
Rome, Italy

Bachelor in Computer Science and Engineering
2017 - 2020

- Final Thesis is the implementation of a social network platform called "Get Together" which uses API and OAuth services to implement its functionalities and provides a RESTful API interface.

Projects

Master Thesis on Continual Learning
Milan, Italy

2024-2025

- Project which aims to counter the effects of **catastrophic forgetting** in deep neural networks through the use of **meta-learning, meta-reinforcement learning and regularization techniques** and bio inspired neuro-modulation techniques in a bio plausible network architecture.
- Technologies: High Permormance Computing infrastructure **(HPC), SLURM, NumPy, Pandas, Ray, PyTorch, Keras, TensorFlow, Matlab, Scikit-Learn, Matplotlib**

Performance analysis
Milan, Italy

Performance analysis and system evaluation using statistical methods in different scenarios
2023

- Implemented through **Matlab** and **Java Modelling Tool (JMT)**.
- Performance indices and measures, workload and service characterization, analytical models, simulation and resource contention.

Image Analysis and Time Series Classification
Milan, Italy

Image Analysis and time series classification on private datasets using different kinds of Artificial Neural Networks models
2022

- The project is implemented through Kaggle's GPUs.
- Technologies: NumPy, Pandas, Ray, PyTorch, Keras, TensorFlow, Matlab, Scikit-Learn, Matplotlib

References

Prof. Annamaria Mesaros Associate Professor of Signal Processing at the Faculty of Information Technology and Communication Sciences at Tampere University.

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